

SELF-SUPERVISED DEEP LEARNING FOR FISHEYE IMAGE RECTIFICATION

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ABSTRACT

To rectify fisheye distortion from a single image, we advance self-supervised learning strategies and propose a unique deep learning model of Fisheye GAN (FE-GAN). Our FE-GAN learns pixel-level distortion flow from sets of fisheye distorted images and distortion-free ones (but not requiring such correspondences), with unique cross-rotation and intra-warping consistency introduced. With such novel self-supervised learning techniques, our FEGAN is able to recover the distortion-free image directly from the single fisheye image input. Our experiments quantitatively and qualitatively confirm the effectiveness and robustness of our proposed model, which performs favorably against recent GAN-based image translation models.

Index Terms— fisheye camera, image rectification, generative adversarial network, deep learning, self-supervised learning

1. INTRODUCTION

Fisheye cameras are widely used in a variety of computer vision tasks. With a large field of view (FOV), the images captured by fisheye cameras contain more content information, favorable for users with particular viewing preferences. Unfortunately, images taken by fisheye cameras often suffer from large geometric distortions, especially around the outer regions of the image. Such undesirable properties might degrade the performance in recognition and segmentation, since the pin-hole camera model assumption cannot be held. Therefore, it is desirable to be able to rectify such artifacts, so that highly structured physical properties of the image content can be properly observed and processed.

Early works on fisheye image rectification often require intrinsic and extrinsic camera parameters for calibration, while such parameters can be estimated from multi-view images [1, 2]. However, it might not be always applicable to collect multi-view images for a scene of interest. Thus, how to address such single-image rectification tasks becomes a more challenging task. Fitzgibbon [3] proposed a division model to approximate fisheye distortion in images with fewer parameters. To estimate the distortion parameters, Bukhari *et al.* [4] detected curved-arcs, while Aleman-Flores

et al. [5] utilized an improved Hough Transform to achieve so. Many works have followed this “plumb-line” assumption to estimate distortion parameters from distorted straight lines [4, 6, 7]. Nevertheless, the above methods are based on the pre-determined hand-crafted low-level features (e.g., edge detection, Hough Transform, etc.), which might have limited abilities in describing image content.

With the recent advances of deep learning, alternative approaches to image rectification based on deep convolutional neural networks (CNNs) have shown promising results. Yin *et al.* [8] predicted distortion parameters, guided by semantic segmentation maps to rectify the fisheye images. Xue *et al.* [9] leveraged CNNs for line segment annotations, and focused on transforming curved arcs into straight lines. Li *et al.* [10] regressed the pixel-wise flow between fisheye and distortion-free images to re-sample the distorted pixels. However, most existing learning-based methods for image rectification require supervision (i.e., ground truth distortion-free images, distortion parameters, line segments, segmentation maps, ground truth flow, or paired image data) [8, 9, 10]. The above assumptions and requirements might not be practical when training deep learning models for real-world applications. Furthermore, all of these works assume that there is only one fisheye distortion model, and thus choose to recover its parameters. In real-life scenarios, however, we may have no prior knowledge about the distorted fisheye images. Assuming the use of a single camera model for image rectification might not be sufficient.

To overcome the above problems, we propose an end-to-end deep learning framework based on Generative Adversarial Networks (GANs) [11], Fisheye GAN (FE-GAN). Our FE-GAN does not require ground truth camera parameters or corresponding distortion-free image during training. By advancing self-supervised learning (SSL) strategy and ideas of optical flow [12], our model implicitly learns intrinsic camera parameters for single-image fisheye rectification.

The contributions of our work are highlighted as follows:

- We are among the first to propose self-supervised deep learning models for fisheye image rectification.
- By utilizing GAN-based models, our proposed FE-GAN advances self-supervised learning (SSL) strate-

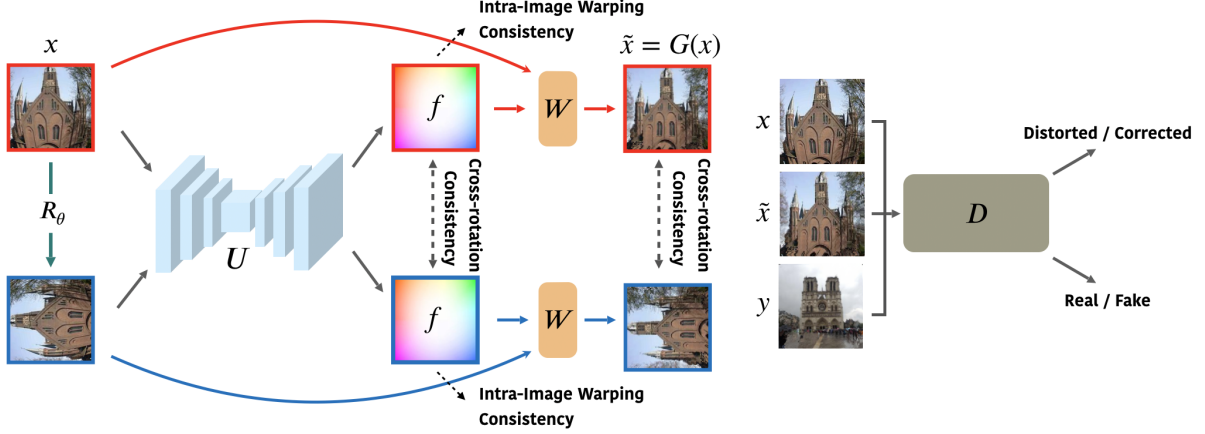


Fig. 1. Network architecture of our proposed Fisheye GAN (FE-GAN), which consists of two major components: a generator $G = (U, W)$ that rectifies the input distorted fisheye image x , and a discriminator $D = (D_{adv}, D_{cls})$. Note that the module U in G predicts the distortion flow $f = U(x)$, and W rectifies the distorted image based on f . In order to realize self-supervised learning, cross-rotation and intra-image warping consistency are uniquely observed as detailed in Sect. 2.2.

gies to observe both image intrinsics and fisheye camera properties for automated rectification.

- Learning of our FE-GAN does not require ground truth camera parameters nor distorted-ground truth image pairs during training.

2. PROPOSED METHOD

2.1. Problem Definition and Notations

To perform single-image rectification for fisheye cameras, we aim at learning a mapping function between the distorted image domain X and the recovered distortion-free image domain Y . During training, unpaired training images $\{x_i\}_{i=1}^N \subset X$ and $\{y_i\}_{i=1}^N \subset Y$ are given. We denote the data distributions as $x \sim \text{data}(x)$ and $y \sim \text{data}(y)$.

Our proposed deep self-supervised learning (SSL) framework of Fisheye GAN (FE-GAN) is illustrated in Fig. 1. Note that an intermediate output is the *distortion flow* $f_i \in F$ of an image x_i , which is an $H \times W \times 2$ feature map (H and W denote the height and width of the input image, respectively). This flow map indicates pixel-wise image coordinate mapping between x_i and the predicted distortion-free image $\tilde{x}_i$. We now detail our network design in the following subsections.

2.2. Fisheye GAN (FE-GAN)

Overall Architecture

As shown in Figure 1, our proposed deep learning framework for fisheye image rectification is based on Generative Adversarial Networks (GANs). Our generator $G : X \rightarrow Y$ aims at recovering the distortion-free images Y from images X captured by fisheye cameras. To be more specific, the genera-

tor part of our network consists of a flow prediction module $U : X \rightarrow F$ and a warping module $W : (X, F) \rightarrow Y$. We note that, the flow prediction module U first takes an input distorted image $x_i \in \mathbb{M}_{H \times W \times 3}$ and outputs a flow map $U(x_i) = f_i \in \mathbb{M}_{H \times W \times 2}$. As for the warping module W , it is applied to recover the rectified image using the predicted flow, and thus the distortion-free predicted output is $\tilde{x}_i = W(x_i, f_i)$.

On the other hand, the discriminator D in our proposed framework consists of two unique modules, i.e., $D = (D_{adv}, D_{cls})$. D_{adv} aims to distinguish between $\{G(x) = \tilde{x}_i\}$ and $\{y\}$, while D_{cls} identifies the output images as distorted ones or those with distortion free. Note that D_{cls} is trained in an adversarial fashion, which views distortion-free images as real and the generated outputs as distorted ones. The objectives of the above adversarial and classification-oriented discriminators are described below:

$$L_{adv}(G, D) = E_{y \sim \text{data}(y)} [\log D_{adv}(y)] + E_{x \sim \text{data}(x)} [\log (1 - D_{adv}(G(x)))], \quad (1)$$

$$L_{cls}(G, D) = E_{y \sim \text{data}(y)} [\log D_{cls}(y)] + E_{x \sim \text{data}(x)} [\log (1 - D_{cls}(x))] + E_{x \sim \text{data}(x)} [\log (1 - D_{cls}(G(x)))]. \quad (2)$$

Cross-Rotation Consistency for SSL

Since SSL does not require any labeled data or those with particular correspondence pairs (e.g., distorted image and its distortion-free version) during training, we now introduce two unique properties, *cross-rotation* and *intra-image warping* consistency, which allow us to realize deep SSL for single-image fisheye rectification.

Inspired by Geometric Consistent Generative Adversarial Networks (GcGAN) [13], we additionally perform

Table 1. Performance comparisons in PSNR, SSIM, and RMSE on image datasets from MCIndoor, Wireframe, and LSUN Churches. Note that the numbers in bold indicate the best results.

Method	MCIndoor			Wireframe			LSUN Churches		
	PSNR↑	SSIM↑	RMSE↓	PSNR↑	SSIM↑	RMSE↓	PSNR↑	SSIM↑	RMSE↓
CycleGAN	11.404	0.335	71.902	11.505	0.257	68.549	10.982	0.193	72.683
UNIT	12.959	0.44	61.387	8.965	0.215	93.236	10.336	0.235	79.812
FE-GAN (ours)	15.215	0.498	45.672	12.846	0.352	59.887	12.781	0.377	59.492

data augmentation by rotating the images by 90, 180, and 270 degrees. We denote such transformation as R_θ , where $\theta \in \{90, 180, 270\}$, and thus enforce *cross-rotation consistency* by rectifying the original distorted input and its rotated version:

$$L_{geo.c}(G) = E_{x \sim data(x)} [\|R_\theta(G(x)) - G(R_\theta(x))\|_1] + E_{x \sim data(x)} [\|G(x) - R_{-\theta}(G(R_\theta(x)))\|_1]. \quad (3)$$

Since the fisheye distortion flow is radial-symmetric, it is expected to be invariant to the rotated versions of the input image, we further calculate the following loss when observing cross-rotation consistency:

$$L_{flow.c}(G) = E_{x \sim data(x)} [\|U(x) - U(R_\theta(x))\|_1]. \quad (4)$$

Intra-Image Warping Consistency for SSL

The above cross-rotation consistency utilizes data-augmentation like techniques for recovering image or distortion flow outputs of interest, without observing the resulting ground truth information. We now advance the intrinsic properties of fish-eye cameras and their images, and enforce another type of consistency for the purpose of self-supervised learning.

With the common assumption that the optical axis goes through the image center, the generated distortion flow f_i of an single image x_i has the following properties:

- *Radial-symmetric*: The distortion flow for each pixel either points outwards or inwards with respect to the image center, while the flow magnitudes for equidistant pixels across the image center remain the same.
- *Smoothness*: Differences in flow magnitudes along horizontal and vertical directions for adjacent pixels would be small.

With the above observations, we have two objectives imposed on $f = U(x)$ centered at $(0, 0)$, while f_p denotes the flow of the pixel at image-coordinate $p \in \mathbb{R}^2$ in the flow map f :

$$L_{smooth}(G) = E_{x \sim data(x)} [\int_f \|\nabla_x f_p\|_1 + \|\nabla_y f_p\|_1], \quad (5)$$

$$L_{radial}(G) = E_{x \sim data(x)} [\int_f \left| \frac{f_p \cdot p}{\|f_p\| \cdot \|p\|} \right| + \int_r Var(\{\|f_p\| \mid \|p\| = r\})]. \quad (6)$$

The full objective $L(G, D)$ sums up the losses in (1) to (6), and thus learning of our model solves the following problem:

$$G = \arg \min_G \max_D L(G, D). \quad (7)$$

3. EXPERIMENTS AND RESULTS

3.1. Datasets

We evaluate our method on three image datasets constructed from MCIndoor20000 [14], Wireframe [15], and LSUN Churches [16]. Following [8, 9, 10], we apply the fisheye transform defined in [17] with randomly determined parameters to synthesize 1000 fish-eye images for each dataset. However, we do not utilize ground truth and distorted image pairs for training.

3.2. Implementation Details

Similar to [12], we utilize cross-connection architectures to assimilate coarse-to-fine generation in our flow prediction module U . Self-attention layers [18] are also used to incorporate global image information, since fisheye distortion is viewed as an image-level characteristic. The warping module W is based on the Spatial Transformer Networks (STN) [19], performing a fully-differentiable re-sampling operation.

Our discriminator $D = (D_{adv}, D_{cls})$ simply consists of multiple down-sampling and strided-convolution layers. Sharing initial layers, D_{adv} outputs real or fake labels while D_{cls} outputs the class labels, i.e. distorted or distortion-free.

3.3. Quantitative Results

We evaluate the effectiveness of our model by comparing with two state-of-the-art image translation methods, CycleGAN [20] and UNIT [21]. As listed in Table 1, we compute PSNR, SSIM, and RMSE between the recovered outputs and the ground truth distortion-free images. From this table, we see that our method performed favorably state-of-the-art unpaired image translation methods across all datasets.

3.4. Qualitative Results

We also perform visual quality comparisons to the aforementioned models of [20, 21] and a popular fisheye rectification

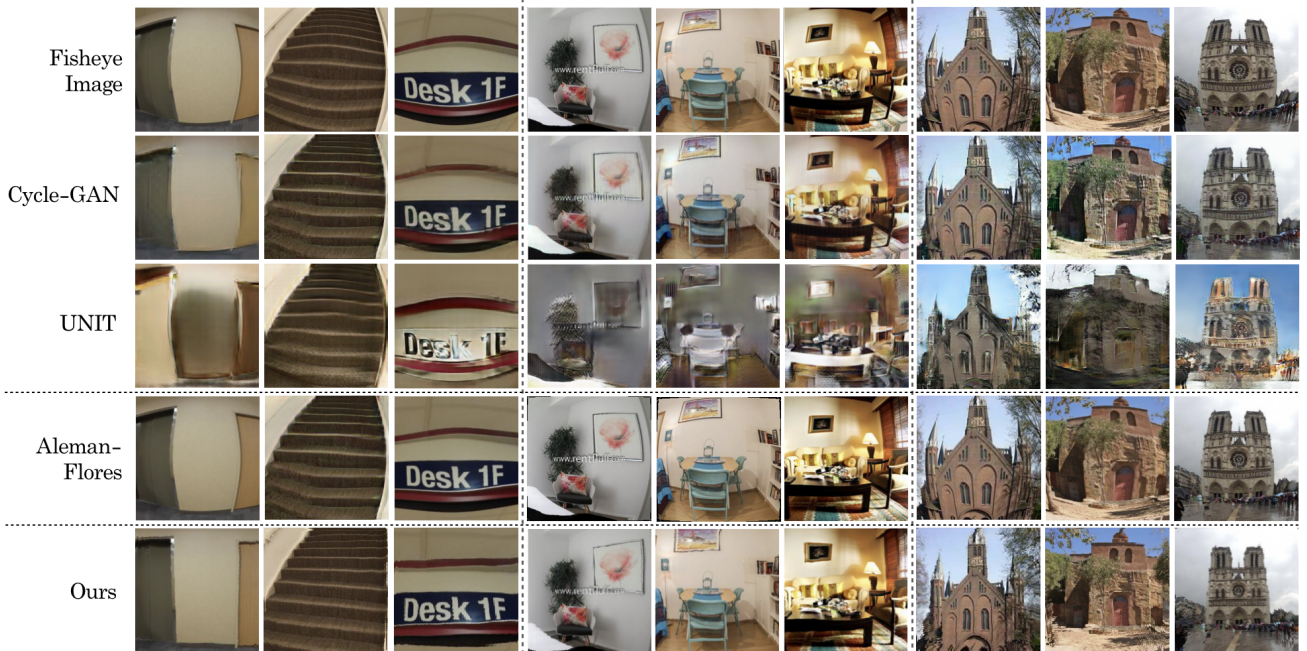


Fig. 2. Example rectification results from MCIndoor, Wireframes, and LSUN Churches datasets. Note that methods of Cycle-GAN [20], UNIT [21], and a popular method by Aleman-Flores [5] are considered for visual comparisons.

method [5] in Fig. 2. From this figure, we see that our method was able to generate visually satisfactory and realistic rectification outputs than the above methods did.

We note that, in contrast with current translation methods that focus on translating image style such as colors and texture, our model is able to observe and incorporate geometric information via the proposed deep learning modules, while content and color information of the input image can be properly preserved. Moreover, as depicted in Fig. 2, our model performed favorably against traditional rectification algorithms by a large margin. This is due to the fact that traditional methods generally assume and use a single distortion model for rectification [3, 5, 4], and thus only regress/predict a fixed number of model parameters.

3.5. Ablation Studies

Finally, we conduct ablation studies to verify the design of our model. As shown in Table 2, we take images from MCIndoor20000 [14] and compare different versions of our model. More precisely, we consider our model excluding self-attention layer, cross-rotation/intra-image warping consistency, and list the results in Table 2. From this table, we see that the full version of our model achieved the best results, which supports the design of our architecture. It is worth repeating that, we do not require supervision in image correspondences (i.e., distorted images and their ground truth versions) during training, and thus is a fully self-supervised learning approach for fisheye image rectification.

Table 2. Ablation studies and performance comparisons of our model using images from MCIndoor [14]. Note that the numbers in bold denote the best performances in terms of each metric.

Method	PSNR	SSIM	RMSE
Ours* (w/o self-attention layer)	12.388	0.361	62.184
Ours* (w/o cross-rotation consistency)	11.021	0.324	72.889
Ours* (w/o intra-image warping)	12.38	0.303	62.319
Ours (full version)	15.215	0.498	45.672

4. CONCLUSION

We presented a unique and novel self-supervised learning model based on GAN, Fisheye GAN (FE-GAN), for rectifying fish-eye distortion from a single image. The novelty of our method lies in the unique design of geometric constraints observed from the input distorted images and the predicted distortion flow (with the resulting output distortion-free images). Using only unpaired images without any ground truth camera parameter supervision or requiring image pairs during training, our experiments confirmed that our model quantitatively outperformed state-of-the-arts image-translation based deep models. More importantly, our FE-GAN was able to produce satisfactory rectified distortion-free images than the aforementioned methods did. Therefore, the effectiveness and robustness of our model can be successfully verified.

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